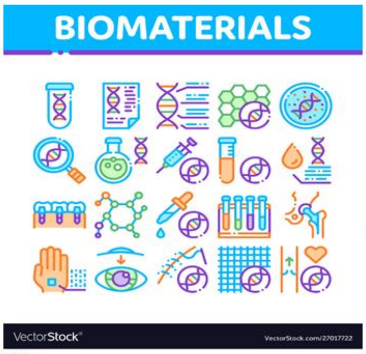
LECTURE-22
(BIOMATERIALS)

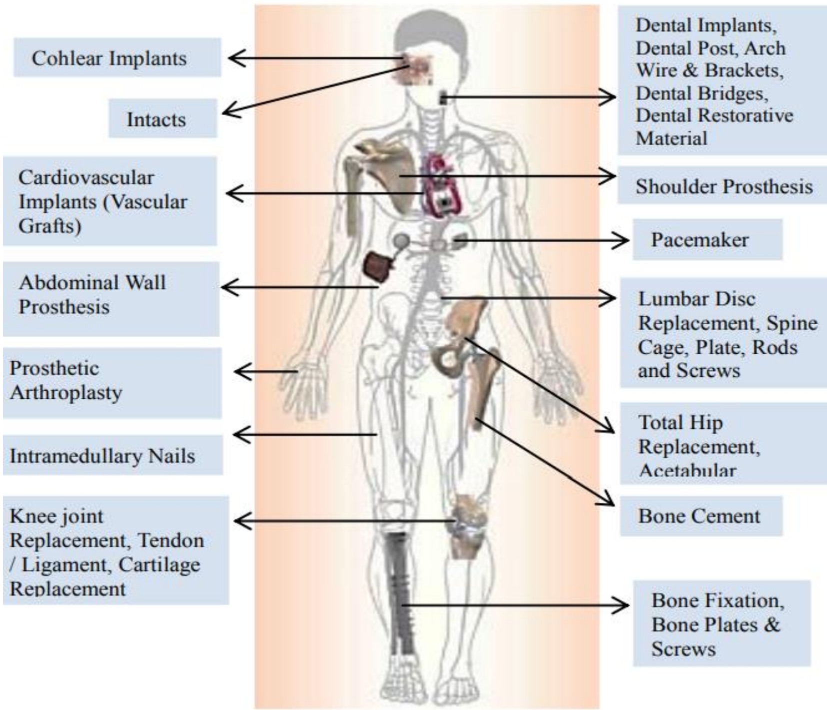
Doctor's hand holding a test tube with contaminated blood sample inside, other hand writing.



Biomaterials Collection

Attractive lab assistant analyzing biomaterial in tubes, biochemical research

(a)



(b)

Implants in the human structure

Example, Hip Implant (c)

Figure-1: Examples of biomaterials based human implants.

➢ INTRODUCTION

- Biomaterial is a synthetic material used to make devices to replace part of a living system or to function in intimate contact with living tissue.
- Biomaterials are materials of metallic, ceramic, polymeric, or composite origin that are used or that have been designed for use in medical devices or in contact with the body in order to augment, replace, or restore the function of diseased or damaged tissues or organs.
- Biomaterials are materials engineered to interact favorably with biological organisms or molecules.
- These materials can be derived from or produced by an organism, or can even be a synthesized polymer.
- Engineers use these novel materials in a wide range of applications, such as tissue engineering, biosensing and drug delivery.
- Applications involving biomaterials range from load bearing orthopedic devices to degradable sutures to living skin equivalents.
- Although the types of materials used as biomaterials is diverse, most implanted biomaterials elicit a very similar inflammatory and foreign body response.
- In an attempt to minimize adverse biological responses and improve the performance of biomaterials at a tissue, cellular, and molecular level, current research in biomaterials has led to a paradigm change in which biomaterials consist of, mimic, or are designed to interact specifically with biological elements and systems.
- Continued adaptation of biomaterials design to biological responses will result in better performing devices that work with the body to promote tissue regeneration and healing.
- Traditionally, they consist of metallic, ceramic, or synthetic polymeric materials, but more recent developments in biomaterials design have attempted to incorporate materials derived from or inspired by biological materials (e.g., silk and collagen).
- The prevalence of biomaterials within society is most evident within medical and dental offices, pharmacies, and hospitals.
- However, the influence of biomaterials has reached into many households with examples ranging from increasingly common news media coverage of medical breakthroughs to the availability of custom color non-corrective contact lenses.
- The evolving character of the discipline of biomaterials is evidenced by how the term biomaterial has been defined.
- In 1974, the Clemson Advisory Board, in response to a request by the World Health Organization (WHO), stated that a biomaterial is a ‘‘systemically pharmacologically inert substance designed for implantation within or incorporation with living tissue’’.
- Dr. Jonathan Black further modified this definition to state that a biomaterial is ‘‘any pharmacologically inert material, viable or nonviable, natural product or manmade, that is part of or is capable of interacting in a beneficial way with a living organism’’.
- National Institute of Health (NIH), USA consensus definition appeared in 1983 and defined biomaterials as ‘‘any substance (other than a drug) or combination of substances, synthetic or natural in origin, which can be used for any period of time, as a whole or as a part of a system that treats, augments, or replaces any tissue, organ, or function of the body’’.
- Thus, relatively newer definitions of the term biomaterial recognize that more modern medical and diagnostic devices will rely increasingly upon direct biological interaction

between biological molecules, cells, and tissues and the materials from which these devices are manufactured.

- These materials, known as biomaterials, include synthetic polymers and, to a lesser extent, biological polymers, metals, and ceramics.
- Specific applications of biomaterials range from high-volume products such as blood bags, syringes, and needles to more challenging implantable devices designed to augment or replace a diseased human organ (Figures-1 and Figure-2).

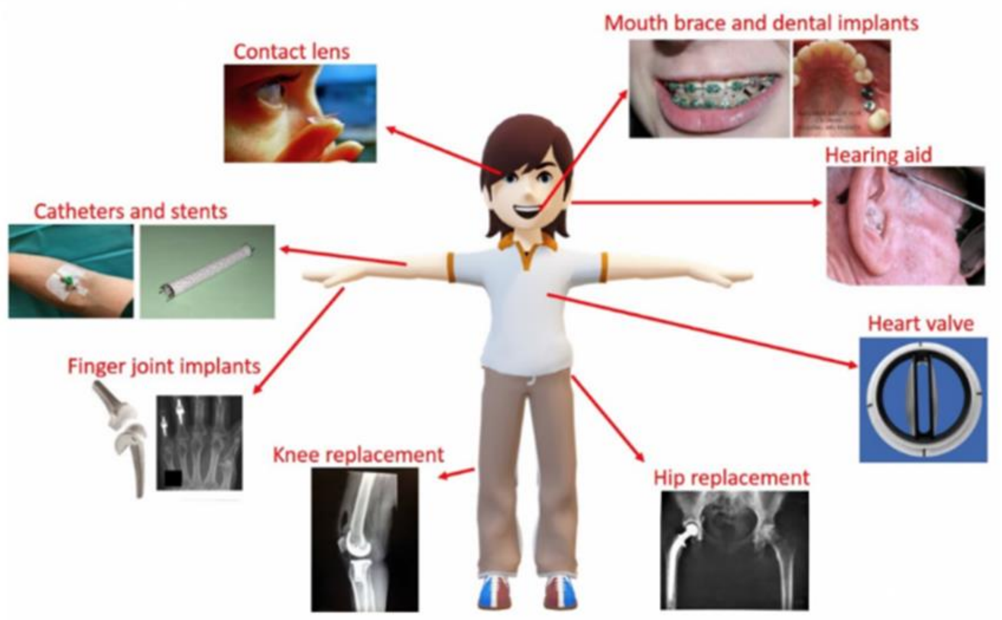


Figure-2: Application and examples.

➢ HISTORY OF BIOMATERIALS

- Compared with the much larger field of materials science, the field of biomaterials is relatively new.
- Although there exist recorded cases of glass eyes and metallic or wooden dental implants (some of which can be dated back to ancient Egypt), the modern age of biomaterials could not have existed without the adoption of aseptic surgical techniques pioneered by Sir Joseph Lister in the mid nineteenth century and indeed, did not fully emerge as an industry or discipline until after the development of synthetic polymers just prior to, during, and following World War II.
- Prior to World War II, implanted biomaterials consisted primarily of metals (e.g., steel, used in pins and plates for bone fixation, joint replacements, and the covering of bone defects).
- n the late 1940s, Harold Ridley observed that shards of poly(methyl methacrylate) (PMMA), from airplane cockpit windshields, embedded within the eyes of World War II aviators did not provoke much of an inflammatory response.
- This observation led not only to the development of PMMA intraocular lenses, but also to greater experimentation of available materials, especially polymers, as biomaterials that could be placed in direct contact with living tissue.
- As the fields of cellular, molecular, and developmental biology began to grow during the 1970s and 1980s, new insights into the organization, function, and properties of biological systems, tissues, and interactions led to a greater understanding of how cells respond to their environment.

- This wealth of biological information allowed the field of biomaterials to undergo a paradigm shift.
- Instead of focusing primarily on replacing an organ or a tissue with a synthetic, usually nondegradable biomaterial, a new branch of biomaterials would attempt to combine biologically active molecules, therapeutics, and motifs into existing and novel biomaterial systems derived from both synthetic and natural sources.
- Although there exist many examples of successful, commercially available biomaterials consisting of metallic and ceramic bases, the focus of biomaterials research has shifted to the development of polymeric or composite materials with biologically sensitive or environmentally controlled properties.
- This change has resulted largely due to the reactivity and variety of chemical moieties that are found in or that can be engineered into natural and synthetic polymers.
- Indeed, by viewing biomaterials as materials designed to interact with biology rather than being inert substances, the field of biomaterials has exploded with innovative designs that promote cell attachment, encapsulation, proliferation, differentiation, migration, and apoptosis, and that allow the biomaterial to polymerize, swell, and degrade under a variety of environmental conditions and biological stimuli.
- Evidence of this polymer and composite revolution is the dramatic increase in the number of publications relating to biomaterials research.
- **Figure-2** shows growth of publications in biomaterials research (1999-2013).

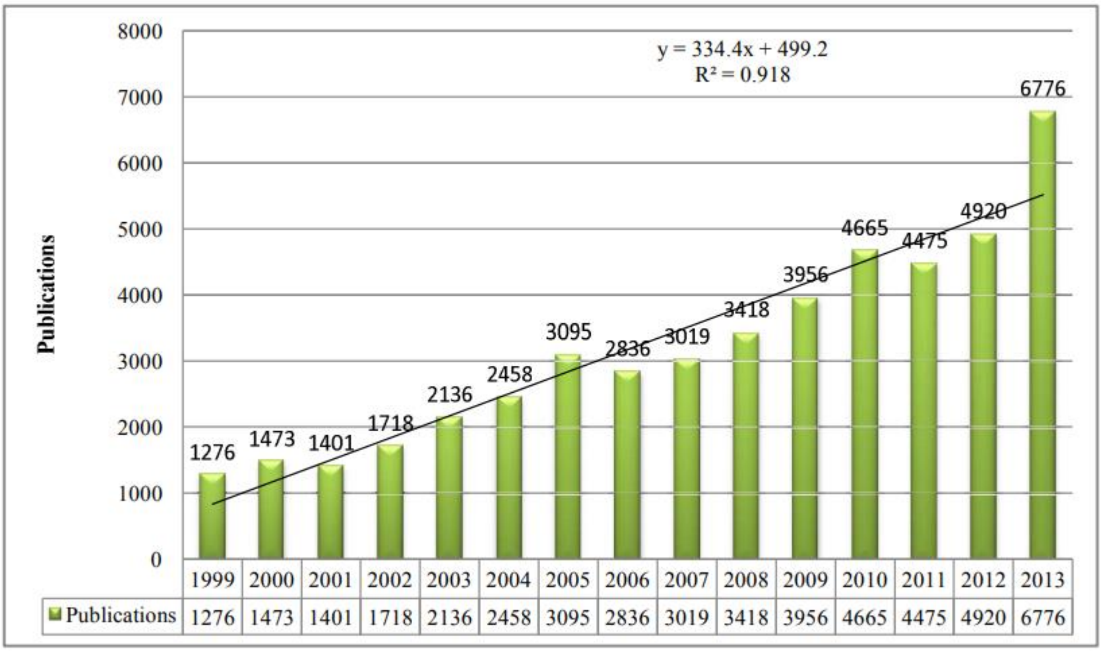


Figure-3: Shows growth of publications in biomaterials research (1999-2013).

- The total number of publications extracted in the field of biomaterials is 47,622 during 1999- 2013 as reflected in Web of Science database and 9, 50,044; citations were received for 47,622 total publications.
- During the period, the exponential growth pattern was reflected by the value of R^2 from WoS (R^2 = 0.918).
- The result confirms the fast growth of publications in biomaterials research during 1999-2013.

APPLICATION OF BIOMATERIALS

- Biomaterials are the kind of materials that are introduced into the body as a medical device for medical purposes.
- These materials are having numerous medical applications such as cancer therapy, artificial ligaments and tendons, orthopaedic for joint replacements, bone plates, and ophthalmic applications in contact lenses, for wound healing in the form of surgical sutures, clips, nerve regeneration, in reproductive therapy as breast implants, etc.
- These materials also have some non-medical applications such as to grow cells in culture medium, assay of blood proteins in laboratories, etc.
- Biomaterials have got thousands of applications, some of them have been summarized in tables-1,2,3

Table-1: Use of Biomaterial

S.No	Problem area	Examples
1	Replacement of diseases or damaged part	Artificial hip joint, kidney dialysis machine
2	Assist in healing	Sutures, bone plates and screws
3	Aid to treatment	Catheters, drains
4	Aid to diagnosis	Probes and catheters
5	Correct cosmetic problem	Augmentation mammoplasty, chin augmentation
6	Correct functional abnormality	Harrington spinal rod
7	Improve function	Cardiac pacemaker, contact lens

Table-2: Organ Examples

Organs	Examples
Bladder	Catheter
Kidney	Kidney dialysis machine
Bone	Bone plate
Ear	Artificial stapes, cosmetic reconstruction of outer ear
Eye	Contact lens, eye lens replacement
Lung	Oxygenator machine
Heart	Cardiac pacemaker, artificial heart valve

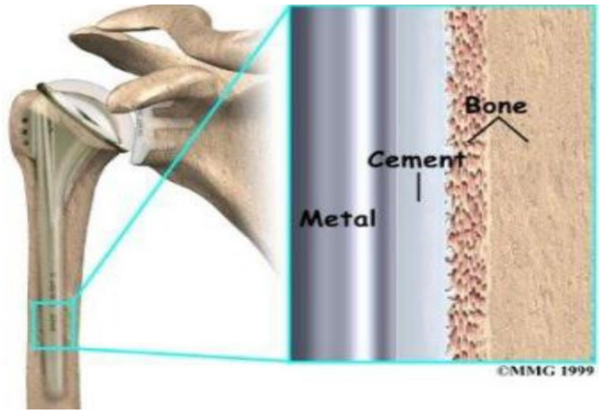
Table-3: Biomaterials in Body Systems

S.No	System	Examples
1	Reproductive	Augmentation mammoplasty, other cosmetic replacements
2	Endocrine	Microencapsulated pancreatic islet cells
3	Nervous	Hydrocephalus drain, cardiac pacemaker
4	Urinary	Catheters, kidney dialysis machine
5	Integumentary	Sutures, burn dressings, artificial skin
6	Respiratory	Oxygenator machine
7	Circulatory	Artificial heart vales, blood vessels
8	Digestive	Sutures
9	Muscular	Sutures
10	Skeletal	Bone plate, total joint replacement

Table-4: Biomaterials for use in the Body

S.No	Materials	Advantages	Disadvantages	Examples
1	Polymers, Nylon, Silicones, Telfon, Dacron	Resilient Easy to fabricate	Not strong Deform with time May degrade	Sutures, blood vessels, hip socket, ear, nose, other soft tissues
2	Metals, Titanium, Stainless steels, Co-Cr Alloys, Gold	Strong, tough Ductile	May corrode Dense	Joint replacement, bone plates and screws, dental root implants
3	Ceramics, Aluminum oxide, Carbon Hydroxyapatite	Very biocompatible, inert Strong in compression	Brittle Difficult to make Not resilient	Dental, hip socket
4	Composites, Carbon-carbon	Strong, tailor-made	Difficult to make	Joint implants; heart valves

- **Figures-4, 5 and 6** depicts the application of biomaterials in orthopedic, dentistry and opthalmic areas of medical sciences.



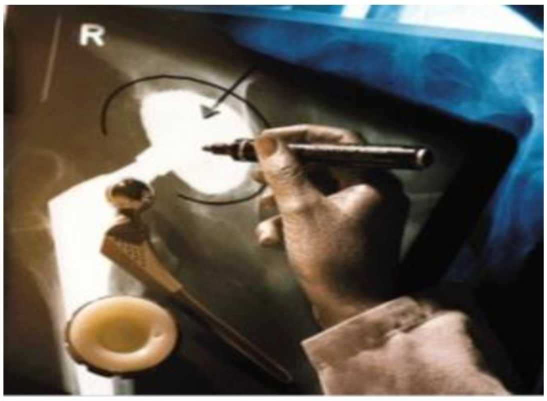


Figure-4: Application in orthopedic.


Figure-5: Application in dentistry.

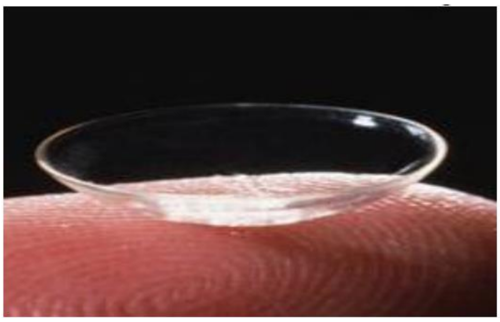
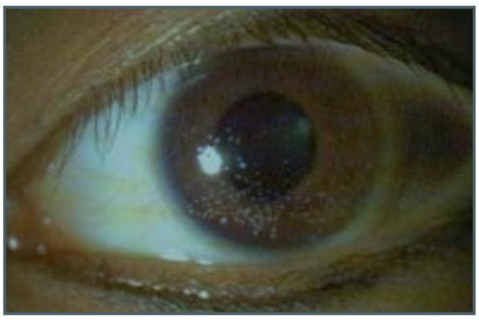



Figure-6: Application in ophthalmic.

- Advancing biomaterials of human origin for therapeutics and tissue engineering have been highlighted by the **Figure-7**.

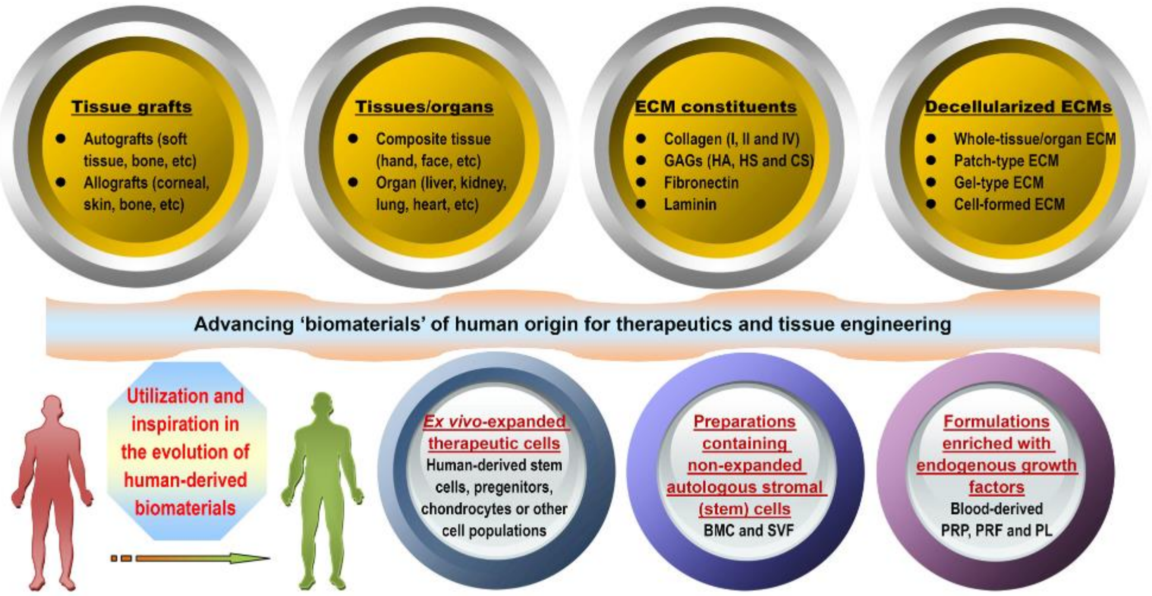


Figure-6: Advancing biomaterials of human origin for therapeutics and tissue engineering.

- **Advancing biomaterials of human origin for tissue engineering has been shown in the Figure-7.**

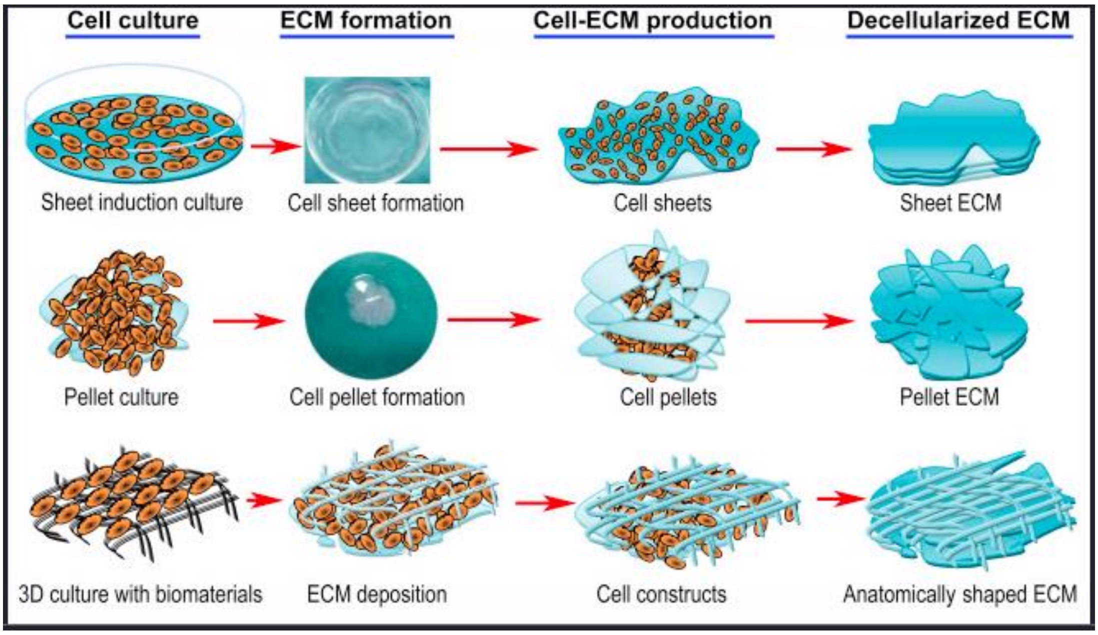
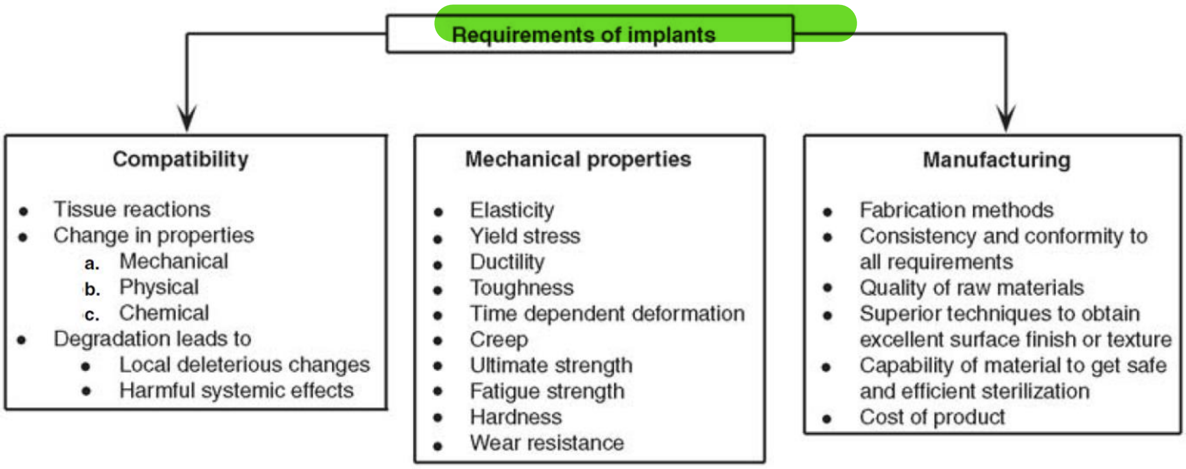


Figure-7: Advancing biomaterials of human origin for tissue engineering.

➢ BIOMATERIAL COMPATIBILITY



- Biocompatibility is related to the behavior of **biomaterials** in various environments under various chemical and physical conditions.
- The term may refer to specific properties of a material without specifying where or how the material is to be used.
- In a simple sense, **materials** are **biocompatible** when they exert the expected beneficial tissue response and clinically relevant performance. The other components of **biocompatibility** are cytotoxicity, genotoxicity, mutagenicity, carcinogenicity and immunogenicity.
- Compatibility of biomaterials have been summarized in the **Figure-8.**

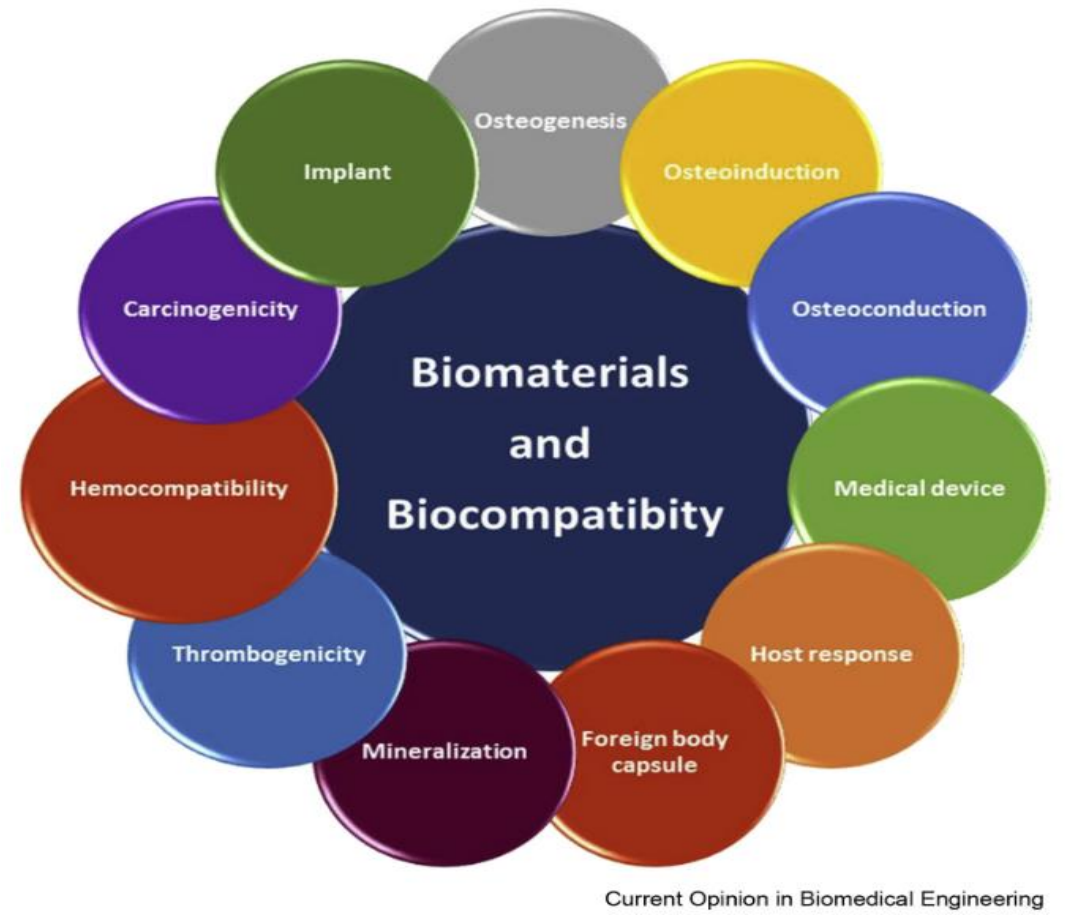


Figure-8: Compatibility of biomaterials.

CHARACTERISTICS AND PROPERTIES

- The design and selection of a biomaterial depend on its specific application in order to be useful and assure its properties as long as required, without rejection.
- In order to be considered a biomaterial it should, besides not inducing inflammation, toxic reactions and allergenic symptoms in the body, be biocompatible, biofunctional, bioactive, bioinert and sterilizable (**Figure-9**).
- A further condition for a biomaterial is that the processes of hygienization and sterilization should not promote modifications in the properties of same.
- The biomaterial surface, being directly exposed to the living organism, plays a crucial role as regards biocompatibility.
- When this issue is considered, the physical, morphological, and biologic characteristics can be adapted to promote improvement in the interaction of biomaterial and tissue.
- The concept of biocompatibility has been deeply altered in these last few years.
- At first it was believed that a material was biocompatible by being completely inert to the human body, without any response from the biological medium to its presence.
- The idea of a totally inert material was abandoned when it could be observed that the presence of any kind of material always entails some response from the body, varying as a function of the kind of application and the patients' characteristics (age, sex, etc.).
- Thus, the concept of biocompatibility of a material, by emphasizing especially the tissue (or neighboring medium) material interface, could be defined only by encompassing the various forms of interaction of the body with the material.
- Characteristics that a material must present to be a biomaterial has been summarized in the **Figure-9**.

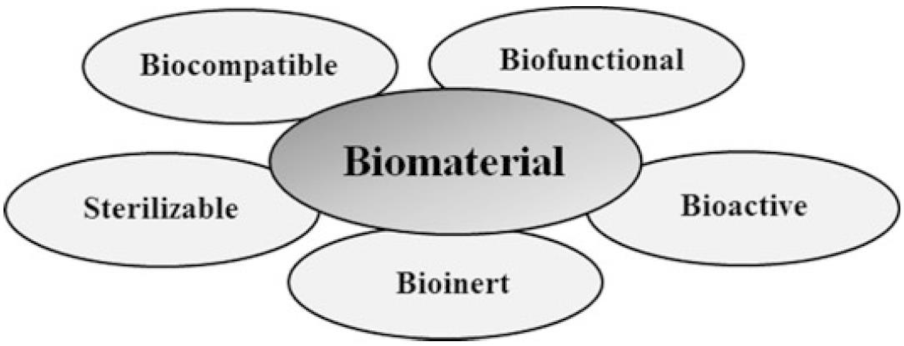


Figure-9: Characteristics that a material must present to be a biomaterial.

- Biocompatibility of a biomaterial as previously defined is related to the organism accepting the body, until then a foreign matter, and admitting it as part of the whole, as well as not exhibiting signs of harmful effects at its locus of operation.
- The expression biocompatibility comprehends different properties of materials, such as toxicity, tissues compatibility, blood compatibility (hemocompatibility), and biofunctionality properties.
- Biofunctionality is essential for the material to perform the functions to which it has been designed for, at the required period of time and effectively.
- For materials of long-lasting biofunctionality the material should not degrade by contact with the tissue; since thus it would partly lose its functionality, it should also keep its chemical and mechanical stability throughout the whole process to which it

has been enabled. For metals, degradation is related to corrosion which causes harmful effects to the body if these are not of a tolerable degree.

- As for the response of the body, biomaterials can be characterized as biotolerable, bioinert, bioactive, and biodegradable.
- Biotolerable is the way by which are designed materials that are tolerated by the body, being separated from the host tissue by the formation of a housing of fibrous tissue.
- This layer is induced by the release of chemical compounds, ions, corrosion products, and other ones resulting from the implanted material.
- Among the biotolerable materials are highlighted the synthetic polymer materials and most of the metals.
- Bioinert are materials which are also tolerated by the body, but where there is no chemical reaction between the material and the tissue.
- The material either does not release components in any significant amount or does not release components and the formation of fibrous housing is minimal. Common examples of bioinert materials are alumina, zirconia, titanium and alloys and carbon.
- Bioactive are materials that can make chemical bonds with the bone tissue, characterized as an osseo integration process.
- Through this process the bone tissue connects to the implanted material promoting the coating by bone cells.
- Materials employed as bioactive are the calcium phosphate, hydroxyapatite, and calcium phosphate compound-based glasses and vitroceramics.
- Biodegradable are materials which degrade, solubilize, or are absorbed when in contact with the body for a certain period of time.
- Their application is of deep interest since there is no need of a further surgery for the withdrawal of the implant.
- Within this class most important are the biopolymers.
- A biomaterial which otherwise complies with all the characteristics required for its utilization can only be effectively employed if it can be sterilized.
- Sterilization is crucial since it is by means of this procedure that all impurities and microorganisms are eliminated, providing a material which is suitable for implant.
- Note: The choice of the biomaterial is associated with the function it will perform and the period of time it will contact the human body, based on the combination of its physical, chemical, mechanical, and biological properties.

PROPERTIES OF BIOMATERIALS

- The biomaterial success for a certain application is related, besides the specific items for biomaterials cited above, to its physical, chemical, and mechanical properties.
- According to the effort demanded from the material and the region to which it will be applied, such properties can be modified and thus they deserve careful attention when choosing a material.
- **Physical Properties**
  - ✓ The physical properties of the biomaterial are fundamental for the response of cell adhesion.

- When cells adhere to the biomaterial surface physical chemical reactions between cell and biomaterial occur, such reactions being influenced by factors such as cell behavior, biomaterial surface properties, and environmental factors.
- The biomaterial surface properties include wettability, filler, roughness, softness, and chemical composition.

- **Chemical Properties**
  - The physical properties, composition, and chemical properties also influence the kind of cell bond and determine the biomaterial chemical stability and reactivity.
  - The corporeal ambience is harsh and may cause corrosion of biomaterials.
  - On account of this fact, the biomaterials' chemical stability becomes a relevant factor as regards biocompatibility.
  - Corrosion products may cause adverse reactions to the implant neighborhood.
  - Body fluids are in balance with specific ions under normal physiological conditions.
  - When a biomaterial is implanted the concentration of these ions increases significantly around it and may cause swelling and pain, besides the fact that the corrosion wastes may migrate to other parts of the body and cause undesirable reactions, both for the tissues and the implant.
  - Corrosion of biomaterials alters not only chemical stability, but also affects the mechanical integrity, with possible premature failure of the material.
  - As with corrosion, the corporeal ambience may cause and/or accelerate the biomaterial degradation.
  - Degradation can also be influenced by sterilization processes to which materials are submitted. When the biomaterial is degraded, modifications occur at the material structure and, consequently, modifications in its properties.
  - Surface functional groups can also influence the biomaterial response, since the surface chemical functionality affects adsorbed protein and cell/protein interactions.
  - Commonly investigated functionalities as relates biomaterials are carboxyl(–COOH), hydroxyl (–OH), amino (–NH_2), and methyl (–CH_3) groups.

- **Mechanical Properties**
  - Advances in engineering and medicine require the development of ever more specialized properties for biomaterials.
  - For a biomaterial utilized for a specific mechanical application, some of the requirements studied are Young's modulus, ductility, tensile strength, yield strength, compression strength and fatigue, and wear debris.
  - These properties are evaluated on account of the fact that the human body has different properties for each tissue.
  - For example, elasticity varies from very soft as is the case of brain tissues (*0.1–1 kPa) up to extremely hard or stiff (30 kPa and above), which is the case for the completely mineralized bone.
  - Biomaterials having Young's modulus close to that of the bone are recommended since they assure uniform tensile distribution and avoid stress shielding after implant placement; high values for yield and compression strength properties

avoid fractures and improve functional stability; ductility is important for modeling the biomaterial formation and for dental biomaterials.

SELECTION OF BIOMEDICAL MATERIALS

- The process of material selection should ideally be for a logical sequence involving:
  1- Analysis of the problem;
  2- Consideration of requirement;
  3- Consideration of available material and their properties leading to:
  4- Choice of material.
- The choice of a specific biomedical material is now determined by consideration of the following:
  1- A proper specification of the desired function for the material;
  2- An accurate characterization of the environment in which it must function, and the effects that environment will have on the properties of the material;
  3- A delineation of the length of time the material must function;
  4- A clear understanding of what is meant by safe for human use.

CLASSIFICATION OF BIOMATERIALS

- The most common classes of materials used as biomedical materials are polymers, metals, and ceramics.
- These three classes are used singly and in combination to form most of the implantation devices available today.

- **Polymers**
  - ✓ There are a large number of polymeric materials that have been used as implants or part of implant systems.
  - ✓ The polymeric systems include acrylics, polyamides, polyesters, polyethylene, polysiloxanes, polyurethane, and a number of reprocessed biological materials.
  - ✓ Some of the applications include the use of membranes of ethylene-vinyl-acetate (EVA) copolymer for controlled release and the use of poly-glycolic acid for use as a resorbable suture material.
  - ✓ Some other typical biomedical polymeric materials applications include: artificial heart, kidney, liver, pancreas, bladder, bone cement, catheters, contact lenses, cornea and eye-lens replacements, external and internal ear repairs, heart valves, cardiac assist devices, implantable pumps, joint replacements, pacemaker, encapsulations, soft-tissue replacement, artificial blood vessels, artificial skin, and sutures.
  - ✓ As bioengineers search for designs of ever increasing capabilities to meet the needs of medical practice, polymeric materials alone and in combination with metals and ceramics are becoming increasingly incorporated into devices used in the body.

- **Metals**
  - ✓ The metallic systems most frequently used in the body are:
    (a) Iron-base alloys of the 316L stainless steel
    (b)Titanium and titanium-base alloys, such as
      (i)Ti-6% Al-4%V, and commercially pure ³ 98.9%
      (ii) Ti-Ni (55% Ni and 45% Ti)
    (c) Cobalt base alloys of four types
      (i) Cr (27-30%), Mo (5-7%), Ni (2-5%)
      (ii) Cr (19-21%), Ni (9-11%), W (14-16%)
      (iii) Cr (18-22%), Fe (4-6%), Ni (15-25%), W (3-4%)
      (iv)Cr (19-20%), Mo (9-10%), Ni (33-37%)
  - ✓ The most commonly used implant metals are the 316L stainless steels, Ti-6%-4%V, and Cobalt base alloys.
  - ✓ Other metal systems being investigated include Cobalt-base alloys and Niobium and shape memory alloys, of which (Ti 45% - 55%Ni) is receiving most attention. Further details of metallic biomedical materials will be given later.

- **Composite Materials**
  - ✓ Composite materials have been extensively used in dentistry and prosthesis designers are now incorporating these materials into other applications.
  - ✓ Typically, a matrix of ultrahigh-molecular-weight polyethylene (UHMWPE) is reinforced with carbon fibers.
  - ✓ These carbon fibers are made by pyrolizing acrylic fibers to obtain oriented graphitic structure of high tensile strength and high modulus of elasticity.
  - ✓ The carbon fibers are 6-15mm in diameter, and they are randomly oriented in the matrix.
  - ✓ In order for the high modulus property of the reinforcing fibers to strengthen the matrix, a sufficient interfacial bond between the fiber and matrix must be achieved during the manufacturing process.
  - ✓ Composites have unique properties and are usually stronger than any of the single materials from which they are made.
  - ✓ Workers in this field have taken advantages of this fact and applied it to some difficult problems where tissue in-growth is necessary.
  - ✓ Examples: Deposited Al_2O_3 onto carbon; Carbon / PTFE; Al_2O_3 / PTFE; PLA-coated Carbon fibers.

- **Ceramics**
  - ✓ The most frequently used ceramic implant materials include aluminum oxides, calcium phosphates, and apatites and graphite.
  - ✓ Glasses have also been developed for medical applications.
  - ✓ The use of ceramics was motivated by:
    (i) their inertness in the body,
    (ii) their formability into a variety of shapes and porosities,
    (iii) their high compressive strength, and
    (iv) some cases their excellent wear characteristics.

✓ Selected applications of ceramics include:
(a) hip prostheses,
(b) artificial knees,
(c) bone grafts,
(d) a variety of tissues in growth related applications in
(d.1) orthopedics
(d.2) dentistry, and
(d.3) heart valves.

➤ CLASSIFICATION OF BIOMATERIALS USED AS SUBSTRATE FOR CELL MODELLING

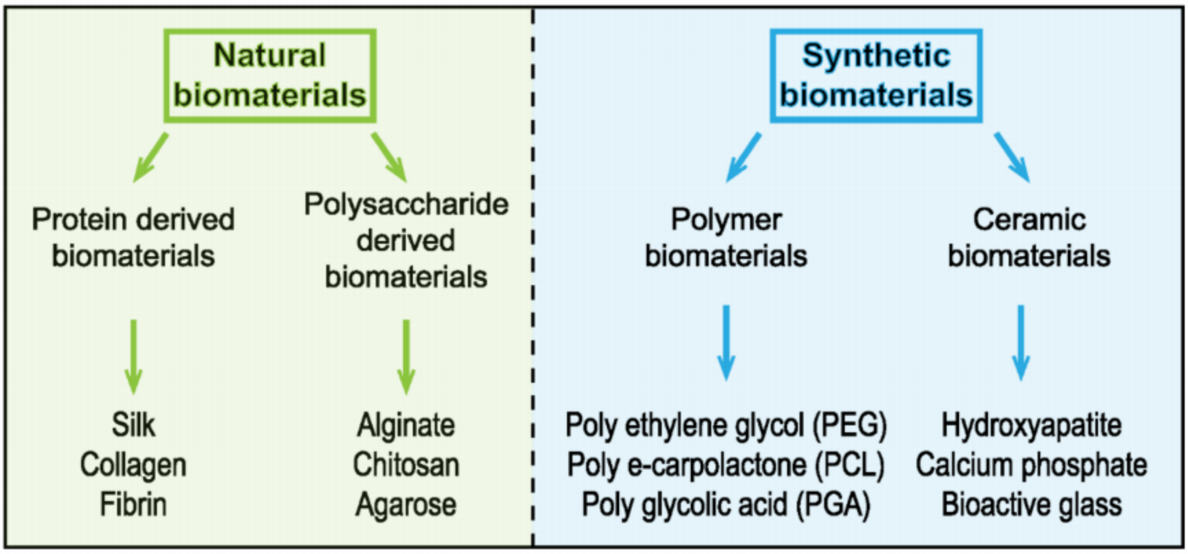
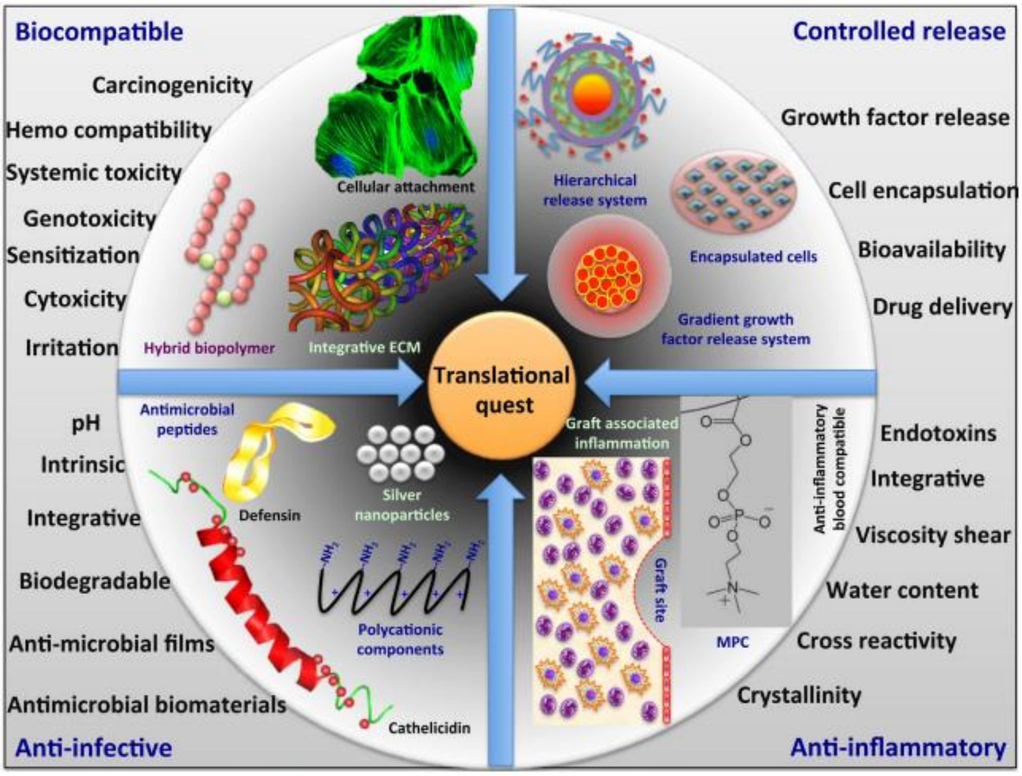
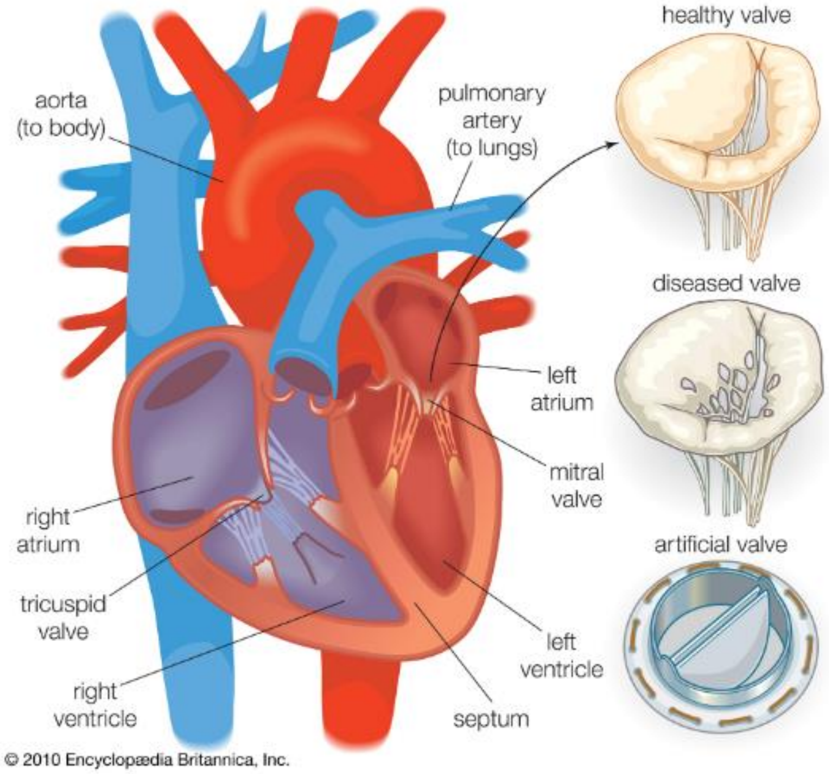


Figure 3. Classification of different biomaterials used as substrate for cell modeling.



Quest for New Generation Biomaterials



Cadiovascular Implant